

Molecular mechanism of signaling of a lysosomal amino acid transceptor

Philémon VON GAUDECKER¹, Layal JAMAL¹, Quentin TAMURA¹, Céline RANSY¹,
Virginie ROPARS¹, Igor TASCON² and José Luis VAZQUEZ-IBAR¹

¹ Intistute of integral biology of the Cell, Gif-sur-Yvette, France; ² Instituto Biofisika, Bilbao, Spain
(corresponding author: philemon.vongaudecker@i2bc.paris-saclay.fr)

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As lysosomes become more and more characterized as signaling hubs, new studies have shown that the lysosomal cationic amino acid transporter PQLC2 [1] is involved in the regulation of an emerging signaling pathway. PQLC2 responds to the cellular cationic amino acid deprivation by recruiting at the lysosomal membrane the C9orf72-SMCR8-WDR41 (or CSW) complex, known for its GAP activity and implication in different lysosomal functions, including autophagy [2]. Pull-down assays in cell cultures indicates that the CSW recruitment occurs via interaction of a protruding flexible loop of WDR41 and PQLC2, where aromatic residues in both proteins seem to play essential roles [3]. To understand PQLC2's amino acid sensing and signaling role, the aim of our project is to characterize the structural elements of PQLC2/WDR41 interaction. For this, we aim to recapitulate in solution the interaction between PQLC2 and WDR41 with the long-term goal to determine the high-resolution structure of the complex.

We engineered a “consensus” sequence of PQLC2, to obtain purified PQLC2 protein with optimal stability in detergent micelles for biophysical and biochemical characterization. Using the engineered PQLC2, we are gathering the first structural information of PQLC2 by Cryo-EM in collaboration with the Biofisika Institute (Bilbao, Spain) where PQLC2 appear to form homo-oligomers. On the other hand, to greatly improve the yield of purification as well as the stability of the WDR41 in solution, we engineered a WDR41 mutant by deleting its C-terminal end - which is known to interact with SMCR8 in the CSW complex. Our stable engineered WDR41 “ΔC” shows specific binding to purified PQLC2 in both pull-down assays and microscale thermophoresis (MST) assays. Competition experiments using synthetic peptides mimicking the WDR41 flexible loop and the use of WDR41 mutants confirmed the essential role of aromatic residues in the loop of WDR41 in mediating WDR41/PQLC2 recognition. Interestingly, we also discovered an unexpected stabilization role of the N-terminal region of WDR41 on PQLC2/WDR41 complex formation. Moreover, replacing four aromatic residues predicted to sit in the cytoplasmic-open cavity of PQLC2 also decreases the affinity between the two proteins. Overall, our results support a model of PQLC2/WDR41 association in which the flexible loop of WDR41 inserts into the cytosolic cavity of PQLC2, while the N-term stabilizes the complex.

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